# Comparison of Linear vs. Deep Learning Dimensionality Reduction Techniques

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## 1 Introduction

Dimensionality reduction is the transformation of data from a high-dimensional space to a low-dimensional space so that the low-dimensional result still retains some important information contained in the original data. This is an important technique dealing with data sparsity as a result of the curse of dimensionality, compressing data to make analysis techniques computationally feasible, and making it possible to visualize data. It is commonly used when analyzing very large datasets or datasets with a large number of variables. PCA finds the orthogonal directions of maximum variance in the data, and classical MDS finds a linear embedding that preserves pairwise Euclidean distances as closely as possible. These methods are computationally efficient and work well for certain types of data, but when non-linear manifolds exist in the high-dimensional data this information may be lost in a resulting low-dimensional structure. We also have nonlinear methods like autoencoders, which are feedforward neural networks that learn a parametric encoder $f_\theta : \mathbb{R}^p \to \mathbb{R}^k$ and decoder $g_\phi : \mathbb{R}^k \to \mathbb{R}^p$ trained jointly to minimize reconstruction error. These methods can be computationally inefficient and can result in overfitting (interpreting noise in the data as important structure).

My research question is: do deep learning-based dimensionality reduction methods (autoencoders) provide a significant improvement in reconstruction error and downstream task performance compared to linear methods (PCA, MDS) across datasets with various structures? To answer this question, I tested all three methods on five datasets:

- A 2D Gaussian point cloud embedded in $\mathbb{R}^{20}$ with Gaussian noise. We expect linear methods to perform well.
- The 3D S-Curve and Swiss Roll. We expect nonlinear methods to perform well.
- The COIL-20 image dataset: 1,440 grayscale images of 20 objects, each photographed at 72 rotation angles. This introduces a real-world circular (nonlinear) manifold embedded in 16,384-dimensional pixel space.
- TCGA BRCA RNA-seq: $n \approx$ 1,200 patients $\times$ 20,530 genes, meaning the number of features greatly exceeds the number of samples.

## 2 Background and Methods

### 2.1 Principal Component Analysis (PCA)

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a centered data matrix. PCA computes the eigendecomposition of the sample covariance matrix $\mathbf{S} = \frac{1}{n-1}\mathbf{X}^\top\mathbf{X}$, which is equivalent to the singular value decomposition $\mathbf{X} =$

$\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. The first $k$ columns of $\mathbf{V}$ form an orthonormal basis $\mathbf{V}_k \in \mathbb{R}^{p\times k}$ for the $k$-dimensional subspace that maximally preserves variance. The encoder is the linear projection $\mathbf{Z} = \mathbf{X}\mathbf{V}_k$ and the decoder is $\widehat{\mathbf{X}} = \mathbf{Z}\mathbf{V}_k^\top$. Among all rank-$k$ linear encoders, PCA achieves the minimum mean squared reconstruction error (Eckart-Young theorem).

### 2.2 Multidimensional Scaling (MDS)

Multidimensional Scaling (MDS) forms a family of methods that map data into a lower-dimensional space by preserving pairwise distances. Classical scaling achieves this by applying a double centering transformation to a distance matrix and extracting its leading eigenvectors. By the Eckart-Young theorem, this yields the best approximation of squared Euclidean distances in a mean squared error sense. Metric Least Squares Scaling generalizes this by finding an embedding $Z_1, \ldots, Z_n \in \mathbb{R}^k$ that minimizes the stress function:

$$\mathcal{L}(Z_1, \ldots, Z_n) = \sum_{i<j} w_{ij}\big(d_{ij} - f(\delta_{ij})\big)^2,$$

where $d_{ij} = \|Z_i - Z_j\|_2$ and $f$ is a fixed monotone function. Because MDS learns an embedding directly rather than a projection mapping from the original space, reconstructing points requires fitting an approximate least-squares linear map $\mathbf{W}$ such that $\mathbf{X}\mathbf{W} \approx \mathbf{Z}$, with decoding performed as $\widehat{\mathbf{X}} = \mathbf{Z}\mathbf{W}^\top$. This method scales poorly in memory and computation, limiting its applicability to larger datasets.

### 2.3 Autoencoders

An autoencoder is a neural network $g_\phi(f_\theta(\mathbf{x}))$ trained to minimize $\|\mathbf{x} - g_\phi(f_\theta(\mathbf{x}))\|_2^2$. The architecture is described by a tuple of hidden dimensions.(256, 64) denotes an encoder $p \to 256 \to 64 \to k$ with ReLU between every linear layer except the output, and a symmetric decoder $k \to 64 \to 256 \to p$. Training uses the Adam optimizer with mean squared error loss. For the TCGA data I added dropout as a regularizer in the hyperparameter grid to guard against overfitting given the small $n$ relative to $p$.

All autoencoder hyperparameters (hidden-layer size, number of layers, learning rate, and dropout rate) are selected by grid search on an 80/20 held-out validation split minimizing validation reconstruction MSE. The final model is then retrained on all data using optimized hyperparameters.

### 2.4 Convolutional Autoencoders

While fully-connected autoencoders treat inputs as flat vectors, this method misses the spatial information contained in image data. To learn local spatial structures, a Convolutional Autoencoder (Conv-AE) is a possible method. The encoder replaces dense layers with a sequence of two-dimensional convolutional layers. For an input image reshaped to a spatial tensor, the encoder applies successive convolutional filters with a 3x3 kernel, a stride of 2, and a padding of 1. The stride of 2 acts as a downsampling mechanism, halving the spatial dimensions at each layer. After passing through ReLU activations, the final spatial feature map is flattened and projected via a dense linear layer to the $k$-dimensional representation.

The decoder acts as the inverse network. It begins with a linear layer that projects the latent representation back to the flattened dimension of the encoder's final feature map, which is then reshaped back into a multi-channel spatial tensor. A sequence of 2D transposed convolutions with a stride of 2 and output padding of 1 upsamples the tensor to reconstruct the original 128x128

image dimensions. ReLU activations are applied after each transposed convolution, except for the final layer, which uses an identity function.

### 2.5 Experimental Design

Reconstruction error is measured in each dataset by reducing the data to 10D or $p-1$ for low-dimensional data where $p$ was the original dimension. Then, the data is decoded back to $p$ and reconstruction error is measured by per-sample MSE:

$$\text{MSE}_i = \frac{1}{p}\sum_{j=1}^{p}\big(x_{ij} - \widehat{x}_{ij}\big)^2.$$

Since each dataset receives 3 MSE values (per method), we can use pairwise Wilcoxon signed-rank tests of the null hypothesis that test if two methods have equal median per-sample reconstruction error. Wilcoxon is used instead of a paired $t$-test because the MSE distributions are strongly right-skewed and not close to normal. I report the test statistic and two-sided $p$-value for every comparison at significance level $\alpha = 0.05$. We also reduce the original datasets to $2D$ for visualization purposes.

For the COIL-20 and TCGA datasets, downstream-task performance is assessed on the 10D embeddings using 5-fold stratified cross-validation. For COIL-20, I use a support vector machine with an RBF kernel ($C = 10$, $\gamma =$ `'scale'`). For TCGA, I use binary logistic regression. The kernel SVM was selected for COIL-20 specifically because logistic regression underperformed for every method. Data are standardized to zero mean and unit variance per feature before any method is applied.

## 3 Synthetic Data Experiments

### 3.1 Datasets

**Linear Subspace (20D).** A 2D Gaussian $Z \sim N(0, I_2)$ with $n = 1000$ points is projected into $\mathbb{R}^{20}$ by a random $2 \times 20$ matrix $\mathbf{A}$, then modified by Gaussian noise ($\sigma = 0.5$). This will make our data lie near a 2-dimensional linear subspace of $\mathbb{R}^{20}$. This is a setting where PCA is expected to be optimal and an autoencoder could hurt by fitting noise.

**S-Curve (3D).** Generated by sklearn, this dataset samples 1000 points from an embedded 2D manifold shaped like the letter "S" in 3D. The manifold is nonlinear but not very complex.

**Swiss Roll (3D).** Generated by sklearn, this dataset is a rolled 2D sheet, which contains points that are nearby in 3D Euclidean space may be far apart on the manifold.

For each synthetic dataset, the points are colored by their latent parameter so that preservation of the manifold structure can be visually assessed.

### 3.2 Results

Table 1: Best autoencoder configuration per synthetic dataset

| Dataset | Hidden Dims | Learning Rate |
|---|---|---|
| Linear Subspace | `(64,)` | $1 \times 10^{-3}$ |
| S-Curve | `(256, 128)` | $1 \times 10^{-3}$ |
| Swiss Roll | `(128, 64)` | $1 \times 10^{-3}$ |

Table 2: Reconstruction MSE on synthetic data. Mean ± standard deviation over 1000 samples.

| Dataset | PCA | MDS | Autoencoder |
|---|---|---|---|
| Linear Subspace (10D) | 0.1341 ± 0.0622 | 0.1700 ± 0.0765 | 0.1344 ± 0.0602 |
| S-Curve (2D) | 0.3121 ± 0.3673 | 0.3019 ± 0.3853 | 0.1191 ± 0.1463 |
| Swiss Roll (2D) | 13.0497 ± 12.1389 | 16.9806 ± 15.2827 | 4.8156 ± 7.1099 |

Table 3: Pairwise Wilcoxon signed-rank tests on per-sample reconstruction MSE

| Dataset | PCA vs. MDS | PCA vs. AE | MDS vs. AE |
|---|---|---|---|
| Linear Subspace | $p < 0.0001$* | $p = 0.2198$ | $p < 0.0001$* |
| S-Curve | $p = 0.2671$ | $p < 0.0001$* | $p < 0.0001$* |
| Swiss Roll | $p < 0.0001$* | $p < 0.0001$* | $p < 0.0001$* |

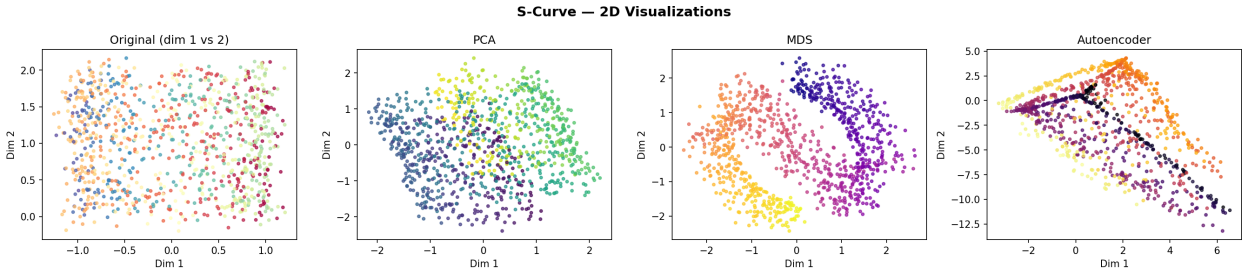



Figure 1: S-Curve dataset: original 3D manifold in the first two dimensions (left), and the 2D embeddings by PCA, MDS, and the autoencoder.

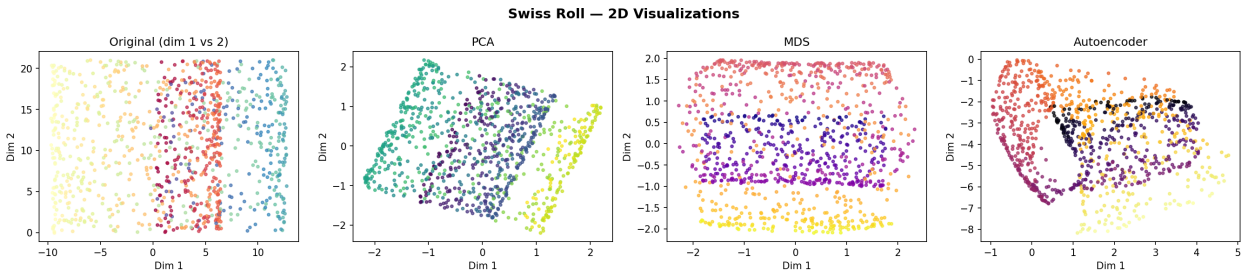



Figure 2: Swiss Roll: 3D manifold and 2D embeddings.

### 3.3 Interpretation

In the Linear Subspace experiment, the true signal lies in a 2D linear subspace of $\mathbb{R}^{20}$. PCA's top 10 components should capture all signal plus some noise and MDS, should match PCA as its iterative solution should approximate the same subspace. An autoencoder with its additional parameters risks fitting noise rather than signal. We see from Table 2 that the autoencoder's mean MSE (0.1344) is very similar to PCA's (0.1341). The autoencoder did not exhibit the overfitting behavior discussed before. This may be because its small single-layer architecture (64 hidden units) selected by validation-set tuning, matched PCA's reconstruction error rather than degrading below it. This underscores the importance of hyperparameter selection for this method.

Classical MDS with Euclidean distances is mathematically equivalent to PCA up to rotation for centered data, so the two methods should produce the same reconstruction MSE. In practice, scikit-learn's SMACOF algorithm for metric MDS is iterative and nonconvex, and the $p < 0.0001$ significant gap between PCA (0.1341) and MDS (0.1700) on this dataset illustrates the practical cost of the method landing at a suboptimal local minimum rather than PCA's closed-form solution. Another source of error could be the approximate back-projection used to reconstruct from MDS embeddings (see Section 4).

The S-Curve and Swiss Roll datasets are intrinsically 3-dimensional but have a 2D manifold, and compressing to 2 latent dimensions requires a method that considers this manifold. The results strongly favor the autoencoder on both datasets. On the S-Curve, the autoencoder achieves a mean MSE of 0.1191 compared to 0.3121 for PCA and 0.3019 for MDS (a 2.6× improvement) with the Wilcoxon tests against both linear methods significant at $p < 0.0001$. On the Swiss Roll, the autoencoder's advantage is similar.

The 2D embedding plots help us to see what these models are doing. A linear method can only flatten the manifold onto a plane, while a successful nonlinear method can unroll it so that the color gradient becomes a clean gradient in the 2D embedding. Figure 2 shows that the autoencoder does successfully unroll the Swiss Roll, producing a more ordered color gradient than either linear method. Interestingly, the MDS method performed on the S-curve found the structure of the data better than either other method. This is impressive, since visualization is an important application of dimensionality reduction beyond reconstruction error.

## 4 COIL-20 Experiments

### 4.1 Dataset and Preprocessing

The Columbia Object Image Library (COIL-20) "processed" version consists of 1,440 grayscale images of 20 everyday objects placed on a motorized turntable and photographed at 72 evenly spaced rotation angles (5° per step). Each image is 128 × 128 pixels, giving the feature space of

a dimension of $p = 16{,}384$. The background has been removed, so all pixel variance is driven by object identity and rotation angle. After loading, pixels are normalized to $[0, 1]$ and standardized to zero mean and unit variance per pixel.

## 4.2 Results

The best fully-connected autoencoder configuration from the hyperparameter search is hidden dimensions `(1024, 256, 64)` with learning rate $1 \times 10^{-3}$ (validation MSE = 0.1785). To see if a different architecture could improve performance on image data, I trained a convolutional autoencoder (Conv-AE) on the same task: four stride-2 convolutional layers reducing $128 \times 128$ images to an $8 \times 8$ feature map, followed by a linear layer to the 10D latent. The best Conv-AE configuration was channels `(32, 64, 128, 256)` with learning rate $1 \times 10^{-3}$ (validation MSE = 0.1556). See appendix for full grids.

Table 4: COIL-20 results summary.

| **Method** | **MSE (10D)** | **Acc. (10D)** |
|---|---|---|
| PCA | 0.0221 | 0.9986 ± 0.0028 |
| MDS | – | 0.9743 ± 0.0089 |
| Autoencoder | 0.0014 | 0.9944 ± 0.0028 |
| Conv-AE | 0.0029 | 0.9993 ± 0.0014 |

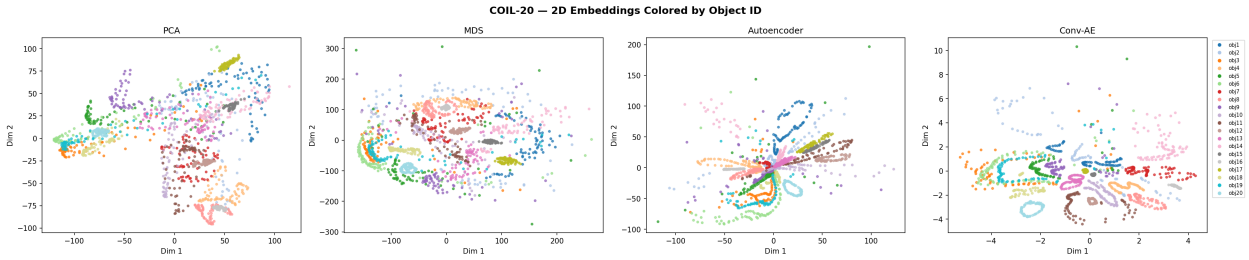



Figure 3: COIL-20: 2D embeddings colored by object ID (20 colors, one per object).

## 4.3 Interpretation

The fully-connected autoencoder achieves a mean reconstruction MSE of 0.0014, roughly 16× lower than PCA's 0.0221, with the Wilcoxon test indicating that every single one of the 1,440 images is reconstructed more accurately by the autoencoder than by PCA. This suggests that the variation in pixel space is nonlinear. The Conv-AE also beats PCA on reconstruction (0.0029 vs. 0.0221), but is roughly 2× worse than the FC autoencoder despite being specifically designed for image data. The FC autoencoder has a direct parameter path between every output pixel and the latent code, while the Conv-AE is constrained to encode spatial features. However, this optimization appears to come at the expense of preserving broader spatial and structural information, which the Conv-AE is constrained to encode.

With an RBF-kernel SVM, all four methods achieve accuracy above 97%, and the Conv-AE takes first place at 99.93% versus PCA's 99.86%, the FC autoencoder's 99.44%, and MDS's 97.43%. The Conv-AE's 0.07-percentage-point lead over PCA is small, but we see from Figure 3 that the Conv-AE separates certain objects far better than any other method. The Conv-AE achieves this

with a higher reconstruction MSE than the FC autoencoder. This shows that reconstruction-MSE has the potential to mislead an analyst about which method is best for a given downstream task.

The mean MDS reconstruction MSE of 13.995 far larger than PCA's 0.022, which is a larger gap than seen in any other dataset. Classical MDS with Euclidean distances is mathematically equivalent to PCA up to rotation, so the true reconstruction error should not differ by such a margin. Since `scikit-learn`'s SMACOF-based MDS returns only an embedding $\mathbf{Z}$ and no inverse map, $\widehat{\mathbf{X}}$ is computed here via a least-squares linear map fit from $\mathbf{X}$ to $\mathbf{Z}$ and then applied in the opposite direction. This is an approximate back-projection which works poorly on a 16,384-dimensional input. Therefore, MSE is left blank in Table 4. Since no inverse map is required for classification accuracy, that entry remains in the table.

# 5 TCGA BRCA Experiments

## 5.1 Dataset and Preprocessing

The TCGA Breast Invasive Carcinoma (BRCA) RNA-seq dataset, downloaded from the UCSC Xena data portal (HiSeqV2), contains $\log_2(x+1)$-transformed gene expression for 20,530 genes across approximately 1,218 samples: roughly 1,097 primary tumor, 114 adjacent-normal, and 7 metastatic samples. Sample type is encoded in the TCGA barcode. To make MDS computationally feasible and to reduce noise from low-expression genes, I restricted analysis to the top 2,000 most variable genes (ranked by sample-wise variance). This leaves a matrix of size $n \approx 1{,}218 \times p = 2{,}000$. Features are standardized prior to any fitting.

With $n \approx 1{,}200$ and hidden layers of a few hundred units, the autoencoder has more parameters than samples and is at genuine risk of overfitting. The hyperparameter grid includes dropout rates of 0 and 0.2 to address this. Gene expression is known to be highly correlated across genes, suggesting a small true intrinsic dimensionality. If most of the signal is captured in a small number of linear principal components, PCA is already near-optimal and an autoencoder has little room to improve.

Figure 4 shows the cumulative explained-variance curve of PCA on the top 2,000 variable genes. The first `14` components explain 50% of the variance, `81` components explain 80%, and `201` components explain 95%.

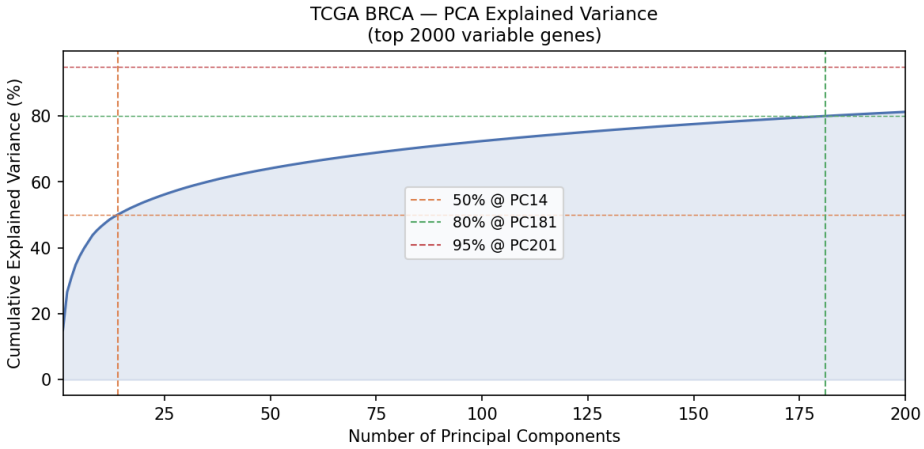



Figure 4: TCGA BRCA: cumulative PCA explained variance across the 2,000 most variable genes. Dashed lines mark the 50%, 80%, and 95% thresholds.

## 5.2 Results

The best autoencoder configuration from the hyperparameter search is hidden dimensions `(256,)`, learning rate $3 \times 10^{-4}$, and dropout 0.2 (validation MSE = 0.523; see appendix and Figure 5 for the full grid). Across the grid, configurations without dropout achieved validation MSE ranging from 0.529 (smallest network) to 0.805 (largest), with the larger architectures showing overfitting, while the variants with a dropout of 0.2 were much more stable (0.52-0.57). This directly confirms that with $n \approx 1{,}200$ samples and hundreds of hidden parameters, regularization was essential. Table 5 summarizes reconstruction MSE and Tumor/Normal classification accuracy.

Table 5: TCGA BRCA results summary. Classification is binary Tumor vs. Normal by 5-fold stratified CV of logistic regression, with Metastatic samples excluded due to low $n = 7$.

| **Method** | **MSE (10D)** | **Accuracy** | **Wilcoxon vs. AE ($p$)** |
|---|---|---|---|
| PCA | 2.827 | $0.9942 \pm 0.0042$ | $< 10^{-80}$ |
| MDS | – | $0.9926 \pm 0.0031$ | $< 10^{-200}$ |
| Autoencoder | 2.399 | $0.9909 \pm 0.0017$ | — |

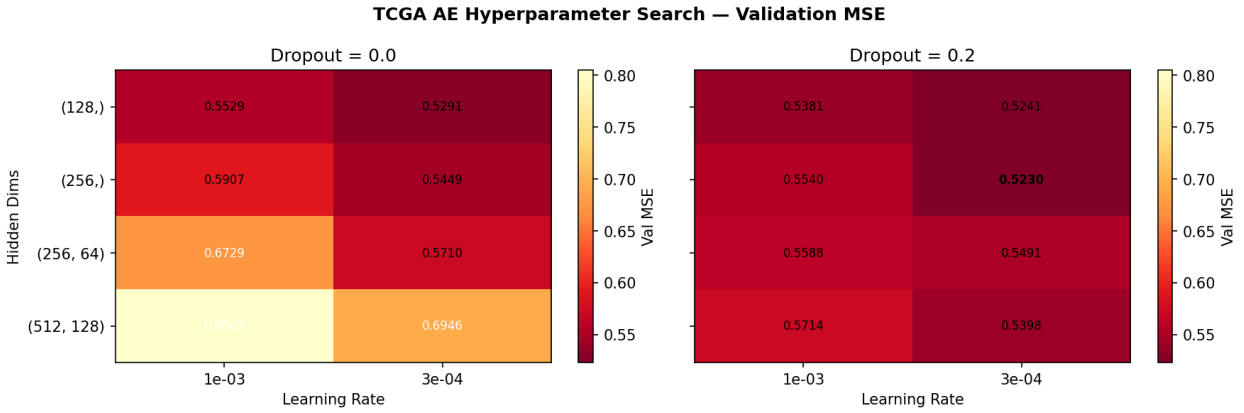



Figure 5: TCGA BRCA autoencoder hyperparameter grid. Each cell is the validation-set reconstruction MSE for a (hidden dims, learning rate, dropout) configuration.

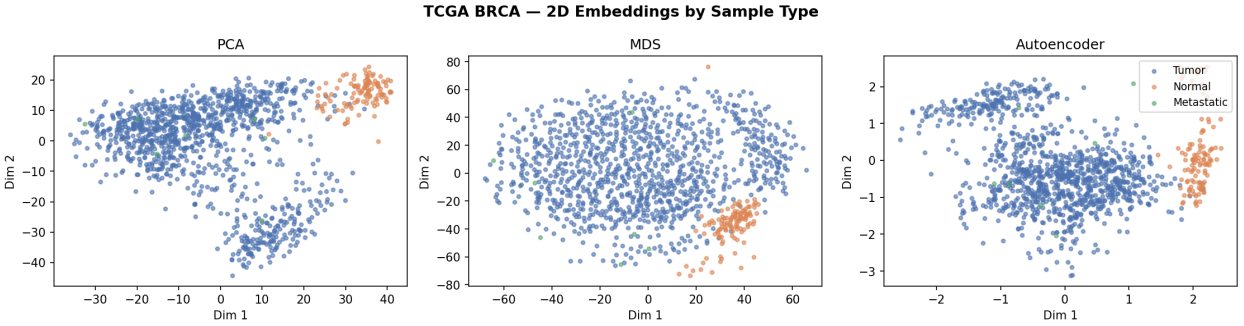



Figure 6: TCGA BRCA: 2D embeddings by each method, colored by sample type.

### 5.3 Interpretation

The autoencoder achieves a mean reconstruction MSE of 2.399 compared to PCA's 2.827, a 15% reduction that is highly statistically significant (Wilcoxon $p < 10^{-80}$). The Wilcoxon statistic is not 0 which indicates that the AE does not beat PCA on every single sample, but the AE's advantage on the remaining majority is overwhelming enough to make the overall difference certain. This suggests that the gene expression manifold is mostly linear, but contains nonlinear structure in some subspaces that the AE is able to exploit when properly regularized.

The hyperparameter search provides evidence for the concern about overfitting risk. Without dropout, validation MSE increased monotonically with network capacity: the single-layer $(128,)$ network without dropout achieved validation MSE of 0.529, while the two-layer $(512, 128)$ network without dropout achieved 0.805 (52% worse despite four times the parameter count). With dropout of 0.2, validation MSE was nearly flat across all architectures (0.52-0.57), indicating that dropout successfully recovered generalization even for the over-parameterized networks.

All three methods achieve Tumor vs. Normal classification accuracy in the 99.0-99.4% range. This suggests that Tumor vs. Normal is a trivially separable classification task in this data. The downstream-task comparison on TCGA is therefore uninformative about method quality. In the future, a more demanding task like molecular subtyping of tumors could provide more information about these method's abilities.

MDS's mean MSE of 1,042 is three orders of magnitude larger than PCA's and reflects the same back-projection artifact discussed in Section 4. As on COIL-20, MDS's embedding-based metric (classification at 99.26%) is reliable, while its reconstruction number is not.

## 6 Discussion

### 6.1 Summary of Findings Across Datasets

Deep learning methods achieved significantly lower reconstruction MSE on every nonlinear dataset tested (S-Curve, Swiss Roll, and COIL-20, all at $p < 0.0001$). However, the COIL-20 experiments reveal a disconnect between reconstruction victory and downstream classification victory. The fully-connected autoencoder achieved the absolute lowest reconstruction MSE (0.0014), yet its 10D embedding yielded the lowest classification accuracy among the top methods (99.44%). The Conv-AE achieved the highest classification accuracy (99.93%) despite having double the reconstruction error of its fully-connected counterpart (0.0029 vs. 0.0014). PCA also outperformed the fully-connected autoencoder in accuracy (99.86%) despite having vastly higher reconstruction error. This leads to the conclusion that evaluating dimensionality reduction techniques based on reconstruction MSE alone risks selecting a suboptimal embedding for the actual downstream task.

The fully-connected autoencoder minimizes pixel-level MSE by optimizing a direct parameter path to every pixel. The Conv-AE, constrained by its spatial kernels and pooling, sacrifices some reconstruction detail to extract structural features, which can be highly separable by an SVM. Meanwhile, PCA performs remarkably well simply because the global directions of greatest variance align with the directions that separate classes in these specific datasets. This suggests that methods should be evaluated directly on task metrics rather than relying on reconstruction error for certain applications.

PCA and MDS are hyperparameter-free and solve deterministically, while autoencoders show substantial variability in performance across architectures, regularizers, and learning rates. The TCGA experiment showed evidence of validation MSE varying wildly across different network capacities, necessitating dropout to prevent overfitting, and the COIL-20 experiments required

separate tuning grids for dense versus convolutional architectures. This illustrates that although deep learning methods can extract manifold structures that simpler methods cannot, they have a higher computational cost and require more effort to perfect. Linear methods remain highly effective as PCA was nearly optimal for the linear subspace and TCGA datasets, and MDS provided the cleanest 2D visual unrolling of the S-Curve manifold.

### 6.2 Limitations

MDS as implemented here is metric MDS with Euclidean distances, which is mathematically equivalent to PCA up to rotation for centered data. Because `scikit-learn`'s SMACOF solver returns only an embedding and no explicit inverse map, reconstruction is approximated here by a least-squares back-projection. This approximation is tight when the forward map is near-orthonormal (as in low-dimensional inputs) but can inflate the reported MSE dramatically in very high-dimensional settings. Other interesting comparisons would include Isomap or locally linear embedding, which use geodesic distances on a neighborhood graph and are more distinct from PCA.

Only one random seed was used for each experiment. Autoencoder training in particular can vary across random initializations. An extension would be to repeat each autoencoder experiment across multiple seeds and report mean and variance.

## 7 Conclusion

This project implemented and compared PCA, MDS, dense autoencoders, and convolutional autoencoders on five datasets of differing structure, using an experimental design based on reconstruction MSE, pairwise Wilcoxon tests, and downstream classification performance. The results show that linear methods and autoencoders excel in different settings characterized by the linearity of the data and the sample-to-feature ratio, autoencoders can recover structural properties that linear methods cannot, and the advantages of deep dimensionality reduction are contingent on careful hyperparameter tuning. The autoencoder is a powerful tool but not a replacement for classical linear methods.

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# A Full Hyperparameter Search Results

For each configuration, validation-set reconstruction MSE at the 10D latent task is reported; the minimum-val-MSE configuration is the one retrained on all data for the main-text results.

**Synthetic datasets.** Autoencoder grid: hidden dims $\in \{(64,), (128, 64), (256, 128), (128, 64, 32)\}$; learning rates $\in$ {1e-3, 3e-4}. Hidden dims are capped to $\max(2p, 4k)$ in `run_all_methods` to avoid over-parameterization on 3D inputs.

**COIL-20.** Autoencoder grid: hidden dims $\in \{(512,), (512, 256), (1024, 256), (512, 128), (1024, 256, 64)\}$; learning rates $\in$ {1e-3, 3e-4}.

**TCGA BRCA.** Autoencoder grid: hidden dims $\in \{(128,), (256,), (256, 64), (512, 128)\}$; learning rates $\in$ {1e-3, 3e-4}; dropout $\in$ {0, 0.2}. See Figure 5 for visualization of the full grid.